

Miscellaneous glycosides (Gentian, Quassia and Discorea)



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Miscellaneous glycosides

Miscellaneous glycosides are broad category of glycosides that consist of sugar molecules (glycons) linked to the non-sugar components(aglycons) through a glycosidic bond. These glycosides do not fit into the more specific classifications such as cardiac glycosides or cyanogenic glycosides, which is why often referred to as '**Miscellaneous.**'

They are characterized by their diverse structures and varied biological activities. These glycosides don't fall neatly into the commonly recognized groups.

Chemistry of miscellaneous glycosides:

- **Miscellaneous glycosides** are a diverse group of compounds with varied **aglycone structures**.
- The **aglycone** can range from **simple aromatic compounds** to **complex steroids**.
- These aglycones are linked to **sugar molecules** via a **glycosidic bond**.
- The **glycosidic bond** can be **hydrolyzed**, splitting the glycoside into **sugar** and **aglycone** components.

- The **biological activity** of miscellaneous glycosides depends on the **structure of the aglycone**.
- They exhibit various effects, including:
 - **Antimicrobial activity**
 - **Anti-inflammatory properties**
 - **Cytotoxic effects**

Properties of miscellaneous glycosides:

- Colorless or having faint colors and have distinctive odor.
- These are soluble in water, alcohol or other organic solvents but solubility can vary depending upon aglycon's nature.
- Miscellaneous glycosides appear as crystals, powders or amorphous solids and have bitter taste.
- Generally stable but can be susceptible to hydrolysis.
- Some miscellaneous glycosides show optical rotation such as salicin.
- Complex miscellaneous glycosides have higher melting and boiling points.
- UV and IR spectroscopy techniques are useful for identifying miscellaneous glycosides.

Extraction of miscellaneous glycosides:

1. The plant material containing glycosides is ground and extracted using a suitable solvent such as water, ethanol or methanol.
2. The choice of solvent depends on the polarity of glycosides and plant materials.
3. After extraction the solution is filtered to remove any insoluble plant material.
4. The solvent is then evaporated to concentrate the glycosides.
5. Finally, the concentrated extract can be purified using techniques like chromatography or crystallization to isolate the desired glycosides.
- 6.

Uses of miscellaneous glycosides:

Medical Uses:

- Used in traditional herbal medicine to treat various health conditions.

- Have potential medicinal applications, such as stimulating appetite and improving digestion.
- Act as antioxidants and possess anti-inflammatory properties.
- Help in pain relief.

Industrial Uses:

- Serve as natural insecticides and pesticides.
- Contribute to plant defense against herbivores and pests.
- Used in the pharmaceutical industry for drug formulation due to their bioactive properties.
- Applied in the cosmetic industry for skincare products due to their antioxidant and anti-aging properties.
- Used as natural sweeteners in the food and beverage industry.

Identification tests for miscellaneous glycosides:

1. Molisch's test
2. Fehling's test
3. Benedict's test.

Gentian:

Gentian (*Gentiana lutea*) is a plant that grows in the mountains in Europe. It's often known as bitter wort or bitter root. Its roots are used as medicine. Gentian contains chemicals that might reduce blood pressure by widening blood vessels.



Common name: Gentian root.

Scientific name: *Gentian lutea*.

Family: Gentianaceae.

Origin: Europe, America, Western Asia.

Part used: Dried rhizome and roots.

Active constituents: Gentiopicrin, gentiamarin, swertiamarin.

Uses:

1. Appetite stimulant
2. Treat digestive disorders
3. Bitter flavoring agent.

Quassia:

It is also known as amargo, bitter-ash, bitter-wood and derived from barks of *Quassia amara* tree native to south America. It has been used for centuries in traditional medicines and as an insecticide.



Common name: Quassia bark, bitterwood.

Scientific name: *Quassia amara*.

Family: Simaroubaceae.

Origin: Tropical South America.

Part used: Dried stem wood.

Active constituents: Quassin, neoquassin, terpenoids.

Uses:

1. Insecticide
2. Bitter flavoring agent
3. Digestive aid
4. Fever reduction.

Discorea:

It includes species commonly known as yams, which are found in tropical and sub tropical regions around the world. The genus includes plants that produce edible tubers which are an important food source in many cultures.



Common name: Yam.

Scientific name: *Dioscorea bulbifera*.

Family: Dioscoraceae.

Origin: Africa, Southern Asia, America.

Part used: Roots.

Active constituents: Diosgenin, yamogenin, glycosides.

Uses:

Used to treat diseases such as:

1. Arthritis
2. Skin diseases
3. Fungal infections
4. Leprosy.